



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI MUTHU MUNIESWARA TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

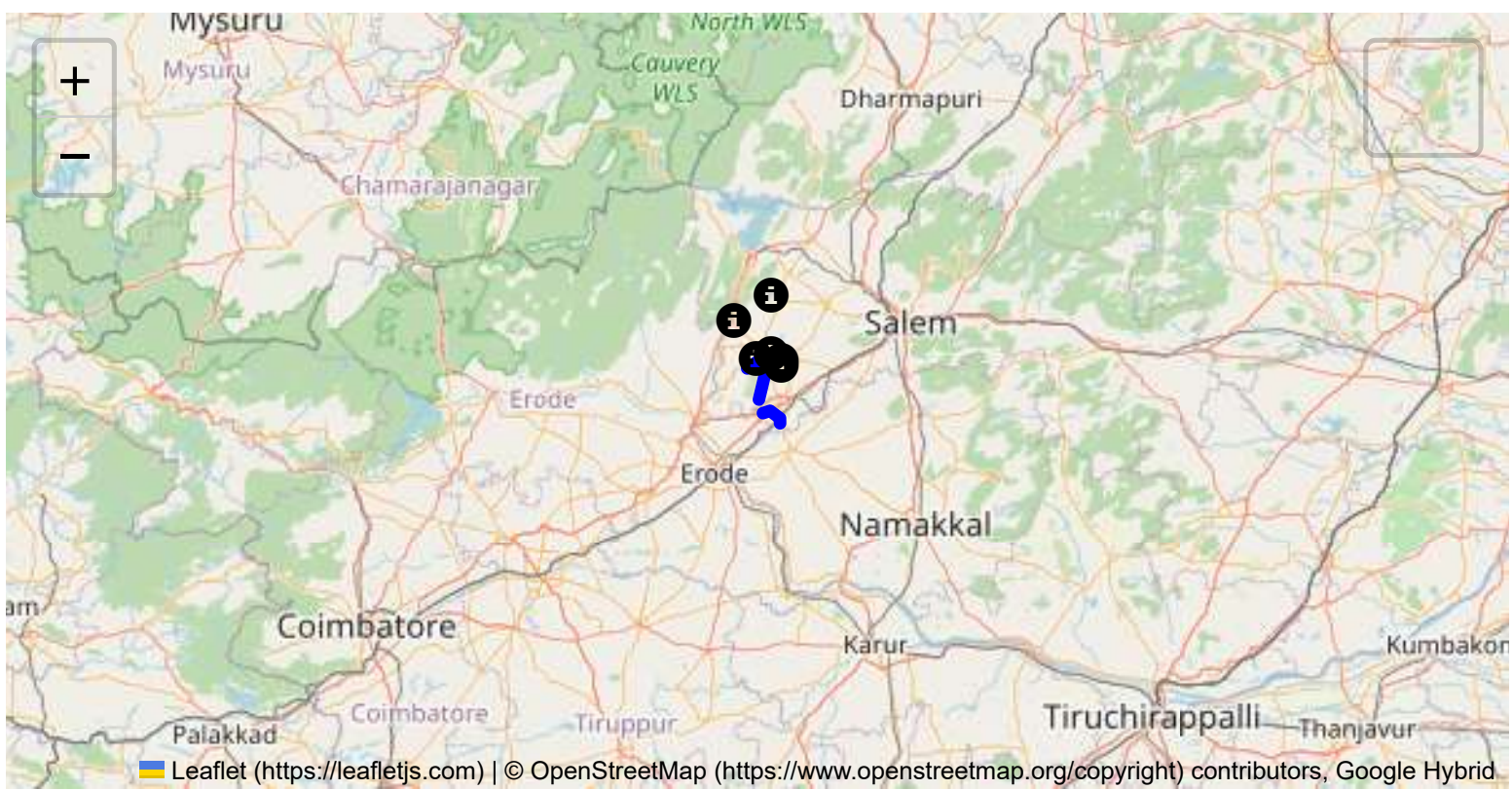
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

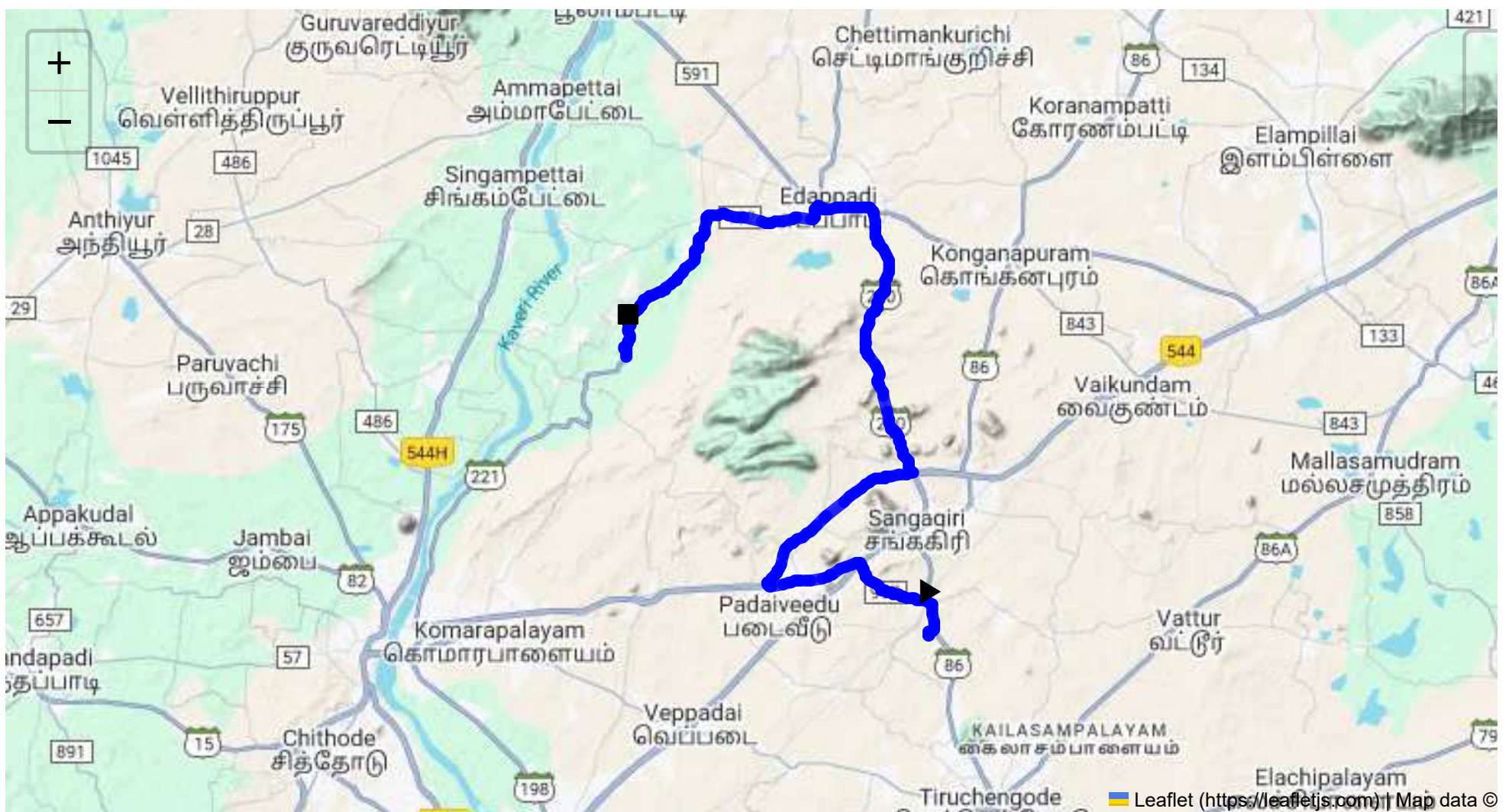
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 40.48 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.533503, 77.767689)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 40.48 kilometers, moving from Sangagiri to Thevur in Tamil Nadu, India. This journey includes major waypoints such as Sankari RS - Erode Privu Rd, Bhavani Main Road (both in Sankari West and Padaiveedu), and Idappadi Rd in Edappadi. These locations suggest a mix of rural roads with some urban intersections, likely featuring state highways and possibly some local roads.

2. Typical Weather Conditions

Tamil Nadu enjoys a tropical climate with hot summers, a monsoon season from October to December, and moderate rainfall. Potential weather-related hazards include:

- Heavy monsoon rains which can lead to slippery roads and flooding.
- Summer heat potentially affecting vehicle performance, especially in June to August.

3. Traffic Patterns

Traffic density can vary:

- Peak traffic likely occurs during morning (7 AM - 10 AM) and evening (5 PM - 8 PM) hours.
- Congestion-prone areas are expected around urban intersections, particularly in and near Sankari and Edappadi.
- Bhavani Main Road may experience increased congestion due to its role as a key thoroughfare.

4. Assessment of Road Quality and Infrastructure

- The main highways like Sankari RS - Erode Privu Rd and Bhavani Main Road are typically maintained but may have variable road quality in some rural sections.
- Infrastructure such as signage and lighting may be lacking in more rural segments.
- Potholes and rough surfaces could be commonplace, especially post-monsoon.

5. Suggestions for Alternative Routes

In case of emergencies or adverse conditions, it is advisable to:

- Utilize alternate state highways, if available, that bypass heavily urbanized areas.
- Consider local feedback for the least disrupted alternate routes during severe weather or closures.

6. Summary of Local Regulations

- Tamil Nadu adheres to the national regulations concerning the transport of hazardous materials, including restrictions on movement during peak city hours and requirements for vehicle compliance.
- Permits must be secured for transporting hazardous materials, and specific routes might be recommended.

7. Historical Incidents Involving Heavy Vehicles

- Historically, some accidents on this route have involved heavy trucks, often due to overloading, speeding, or poor road conditions during monsoons.
- There have been few incidents specific to hazardous materials, but vigilance is advised.

8. Environmental Considerations and Sensitive Areas

- The route passes through agricultural zones, requiring careful management to avoid contamination.
- Any spills or emissions must be promptly addressed to prevent damage to local ecosystems.

9. Communication Coverage

- Mobile network coverage should generally be good along major roads but may drop in remote, rural areas.
- The use of dual SIMs or wireless communication devices is suggested to mitigate dead zone issues.

10. Estimated Emergency Response Times

- Near populated urban areas like Sankari and Edappadi, emergency response times could range from 20-30 minutes.
- Rural segments may experience longer response times, potentially 45 minutes to an hour.

11. Overall Summary of Risk Assessment

This route presents moderate risks typical of regional roads in Tamil Nadu, with particular concerns regarding:

- Weather impacts during monsoon.
- Variable road quality affecting heavy vehicles.
- Possible communication outages in rural sections. Mitigation strategies include using GPS and communication tools proactively, adhering to safe driving practices, especially during adverse weather, and considering alternative routes in event of road closures. Transporters of hazardous materials must remain cautious, adhering strictly to regulatory guidelines, and maintain vigilance across sensitive agricultural zones.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58435, 77.83651	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.46048, 77.85036	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45634, 77.81673	15 KM/Hr
11	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
12	Turn	Medium	11.49404, 77.86724	30 KM/Hr
13	Turn	Medium	11.50866, 77.86053	30 KM/Hr
14	Turn	Medium	11.55039, 77.85511	30 KM/Hr
15	Turn	Medium	11.55460, 77.85624	30 KM/Hr
16	Turn	Medium	11.58508, 77.85281	30 KM/Hr
17	Turn	High	11.58532, 77.83454	15 KM/Hr
18	Turn	Medium	11.58165, 77.83306	30 KM/Hr
19	Turn	Medium	11.58089, 77.83231	30 KM/Hr
20	Turn	High	11.58180, 77.79526	15 KM/Hr
21	Turn	Medium	11.57111, 77.79256	30 KM/Hr
22	Turn	Medium	11.57012, 77.79143	30 KM/Hr
23	Turn	Medium	11.55029, 77.77162	30 KM/Hr
24	Turn	High	11.54528, 77.76829	15 KM/Hr
25	Turn	High	11.54454, 77.76894	15 KM/Hr
26	Turn	Medium	11.54350, 77.76802	30 KM/Hr
27	Turn	Medium	11.54319, 77.76792	30 KM/Hr
28	Turn	Medium	11.53685, 77.76700	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	KRS Hospital and Diabetic Centre	11.5848828, 77.8449664	30 km/h	Medium
2	hospital	Kannan Hospital	11.5854802, 77.8425936	30 km/h	Medium
3	hospital	Aravind Hospital	11.5854615, 77.8424165	30 km/h	Medium
4	hospital	Muthu Eyecare Hospital	11.584736, 77.8393794	30 km/h	Medium
5	clinic	Uma Clinic	11.58398, 77.83977	30 km/h	Medium
6	clinic	KGP Clinic	11.5847763, 77.8387488	30 km/h	Medium
7	hospital	Government Hospital, Edapadi	11.5848264, 77.8380174	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Sri Sankar Hospital	11.5862984, 77.8355021	30 km/h	Medium
9	hospital	Sri Advika Hospital	11.585049, 77.834422	30 km/h	Medium
10	clinic	Xcellent Homecare Clinic	11.584762, 77.8338085	30 km/h	Medium
11	clinic	Raj Physiotherapy Clinic	11.5857357, 77.8341218	30 km/h	Medium
12	hospital	Dr. O.S. Subramanium Hospital	11.584996, 77.834235	30 km/h	Medium
13	hospital	Aravinthu Eye Hospital	11.5866631, 77.8341465	30 km/h	Medium
14	hospital	Mohan Hospital	11.5841331, 77.8336277	30 km/h	Medium
15	hospital	SKM Hospital	11.5841611, 77.8334397	30 km/h	Medium
16	hospital	Nivetha Hospital	11.5858509, 77.833451	30 km/h	Medium
17	clinic	Krishna Hospital and Clinic	11.5846542, 77.8333239	30 km/h	Medium
18	hospital	Rathanasamy Hospital	11.58583, 77.833443	30 km/h	Medium
19	hospital	Pachiappa Hospital	11.58571, 77.833131	30 km/h	Medium
20	hospital	Arunagiri Hospital	11.5837095, 77.8332129	30 km/h	Medium
21	clinic	Rakshan Polyclinic	11.5809367, 77.8326223	30 km/h	Medium
22	hospital	Siddha Hospital	11.5827668, 77.8022539	30 km/h	Medium
23	hospital	Government Hospital	11.563699, 77.7881936	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58435, 77.83651



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49404, 77.86724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58508, 77.85281



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58532, 77.83454



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58165, 77.83306



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58089, 77.83231



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58180, 77.79526



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57111, 77.79256



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57012, 77.79143



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55029, 77.77162



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.54528, 77.76829



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.54454, 77.76894



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54350, 77.76802



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54319, 77.76792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53685, 77.76700
